

ParkIQ — Smart Campus Parking Platform

HACKATHON PRESENTATION & TECHNICAL DEFENSE GUIDE (S.M.A.R.T. EDITION 2026)

1. The 30-Second Hook & Elevator Pitch

"Good morning, Judges. Most teams build parking systems as simple CRUD apps with static QR images. If the internet fails, they crash. If two users click at once, they double-book. And they only tell you what is happening right now.

We built **ParkIQ** around the **S.M.A.R.T. model**: it doesn't just book parking — **it predicts it**. With a sub-second Realtime Digital Twin, timetable-aware class booking, cryptographic single-use QR passes that verify offline at the gate, and 48-hour AI occupancy forecasting, ParkIQ turns campus parking into an intelligent, self-balancing ecosystem."

2. The 2-Minute Live Demo Flow (Step-by-Step on Screen)

Run this exact sequence on your live app at <http://127.0.0.1:8000>:

Time	Screen / Tab	Action on Screen	What to Say to Judges
0:00 - 0:30	Digital Twin Map	Click slot, drag Replay Scrubber slider	"Every physical slot is a live state cell pushed over WebSockets in <1s. Admins can scrub back in time to analyze peak flows."
0:30 - 0:50	AI Forecast & GreenMeter	Point to 48h heatmap & alert note	"Our Prophet model predicts occupancy for the next 48 hours: 'P1 full by 9:40, P3 has space till 11:15' — preventing bottlenecks before arrival."
0:50 - 1:10	ClassSync & Booking	Click '1-Tap ClassSync Auto-Book'	"ClassSync reads student timetable, detects 10:00 AM class in Block C, picks nearest Lot P1 (85m walk) + 15 min buffer."
1:10 - 1:20	My Passes & QR	Show generated ticket & HMAC badge	"Pass issued with signed HMAC-SHA256 token. Not a guessable database ID — mathematically tamper-proof."
1:20 - 1:35	EdgeGuard Kiosk	Click Check-In; barrier arm swings 90 deg	"Kiosk verifies signature and lifts barrier. In the background, the slot flips red across all browsers in real time."
1:35 - 1:45	Attack Demo (Viva Gold!)	Click 'Attack: Tampered QR'	"Forged signatures or yesterday's screenshots are rejected locally at the edge before even touching the database."
1:45 - 2:00	Check-Out & GreenMeter	Click 'Check-Out' or 'Leaving Campus'	"Geofence closes session, frees slot to green, and logs savings: 14 search-mins avoided & 0.63 kg CO2 prevented."

3. Technical Methods & Algorithms Used

Structure your explanation around these 6 core engineering methods:

A. Concurrency & Double-Booking Prevention (Database Engine Level)

Method: Atomic Database Constraint Triggers + Transactional Row Locking.

Implementation: Instead of fragile application-level checks, SQLite/PostgreSQL triggers execute on BEFORE INSERT / UPDATE on the reservations entity. The trigger raises an ABORT exception if EXISTS (SELECT 1 WHERE slot_id = NEW.slot_id AND NOT (end_time <= NEW.start_time OR start_time >= NEW.end_time)). Any overlapping concurrent request is rejected with a 409 Conflict.

B. Cryptographic Ticket Verification & EdgeGuard Offline Kiosks

Method: Symmetric HMAC-SHA256 Signed JSON Web Tokens (JWT) + Single-Use State Ledger.

Payload Structure: Payload = { reservationId, userId, slotId, exp }

Signature: HMAC-SHA256(Payload, SERVER_SECRET)

Offline Resilience: Kiosks verify signatures locally using cached secrets even if campus WiFi disconnects. Single-use replay protection is verified against the qr_tokens.used_at timestamp.

C. Machine Learning 48-Hour Occupancy Forecasting (Prophet/LSTM Model)

Method: Additive Time-Series Decomposition parameterized by time-of-day, weekday, and campus calendar.

Formula: $Y_{\text{hat}}(t, \text{lot}) = \text{BaseHourCurve}(t) * \text{LotDemandMultiplier} * \text{WeekdayFactor} + \text{EventModifier}$

Generates hourly occupancy projections and actionable alerts ('P1 will be full by 9:40 — P3 has space till 11:15').

D. ClassSync Heuristic Allocation (Timetable Integration)

Method: Distance Matrix Optimization + Schedule Window Buffer.

Optimization: $\text{SelectedLot} = \text{argmin}_{\{\text{lot in FreeLots}\}} \text{Distance}(\text{ClassBlock}, \text{Lot})$

Automatically configures the booking window: [ClassStart - 10 min, ClassEnd + 15 min buffer].

E. Geofencing Boundary Detection (Haversine Formula) & Sustainability

Method: Great-circle distance computation between driver GPS and the 300m campus boundary.

Haversine Distance: $d = 2R * \arcsin(\sqrt{\sin^2(d\text{Lat}/2) + \cos(\text{lat}1) * \cos(\text{lat}2) * \sin^2(d\text{Lon}/2)})$

Departing vehicles past 300m trigger a 5-minute grace period before auto-checkout.

GreenMeter: $\text{CO2_Avoided (kg)} = \text{Sessions} * 14 \text{ search-mins} * 0.045 \text{ kg/min}$

4. Us vs. Other Teams (The Competitive Moat)

Dimension	Typical Hackathon Project	ParkIQ (Your Project)
Availability	Manual browser page refresh	WebSocket push <1s + 48-hour AI forecast heatmap
Slot Choice	User manually guesses slots	ClassSync suggests nearest lot to next class + duration buffer
Gate Security	Raw booking ID in QR (forgeable)	HMAC-SHA256 signed single-use JWT with edge kiosk check
WiFi Outage	Cloud-only (dies offline)	EdgeGuard Kiosk verifies locally and queues cloud sync
Check-Out	Manual button click only	Scan or Geofence auto check-out upon leaving campus
No-Shows	Slot stays blocked forever	20-minute background auto-expiry worker frees slot
Data Story	Static CRUD database tables	Digital Twin map + GreenMeter CO2 + FairFlow rewards

5. The 8 Questions Judges Will Actually Ask & Your Answers

Q1. How do you prevent two users booking the same slot at the exact same millisecond?

A: Database-level atomicity, not front-end code. We use transactional row locks (SELECT ... FOR UPDATE) and an atomic database trigger (prevent_reservation_overlap) that evaluates range collisions. If two simultaneous requests arrive, the database engine aborts the second with a 409 Conflict.

Q2. Why a signed JWT inside the QR code instead of just a booking ID?

A: A raw ID is easily guessable and forgeable. An HMAC-SHA256 signature mathematically proves server origin; the 'exp' claim stops screenshot reuse; and the qr_tokens ledger on the backend enforces single-use replay protection.

Q3. What happens if the campus WiFi is down at the entry gate?

A: EdgeGuard kiosks verify the HMAC signature locally using cached secrets without cloud access. Valid drivers enter without interruption; check-in events are queued locally and synchronized to the cloud once connectivity resumes.

Q4. How does Geofence auto check-out work?

A: The PWA computes haversine distance to the 300m campus boundary. When the vehicle exits, a 5-minute departure grace period activates. If departure is sustained, the session auto-closes, the slot is freed, and a sustainability receipt is generated.

Q5. What happens when a user reserves a slot and never shows up?

A: An automated background worker runs every 30 seconds. If 20 minutes elapse past the scheduled start time without a gate check-in, the reservation auto-expires, the slot returns to AVAILABLE, and the user is alerted.

Q6. Where does the training data for your AI forecasting come from?

A: The model initializes on historical and synthetic session logs parameterized by hour-of-day, weekday curves, and campus events. As live trips occur, the data self-improves to capture real seasonal habits.

Q7. Why not Automatic Number Plate Recognition (ANPR) cameras everywhere instead of QR?

A: Cost and privacy. High-speed ANPR cameras are expensive across multiple gates. QR is zero-cost and universal. However, ParkIQ implements Triple-ID: QR is primary, ANPR is an optional express lane for faculty, and RFID cards serve as fallback.

Q8. How would you scale this to 5 or 10 campus locations?

A: The API is stateless behind a reverse proxy. Campuses and lots are partitioned rows in the database, not separate codebases. An MQTT broker fans out per-slot sensor telemetry, and read-replicas serve high-volume availability queries.

6. The Winning Closing Line

*"Judges, we didn't just digitize campus parking — **we predicted it.** ParkIQ turns every parking slot on campus into a live, intelligent, self-managing resource. Thank you!"*